

ETHAN MAYNARD PINTO

Arlington, TX

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Education

University of Texas at Arlington

2022 – 2026

Bachelor of Engineering in Computer Engineering

Arlington, TX

- Relevant Coursework: Embedded Systems, Operating Systems, Digital Logic Design, Electronics, Signal Processing, System on Chip, RISC Processor Design, Data Structures and Algorithms, Cybersecurity Engineering

Experience

Electronic Control Ind. Services

Jun 2024 – Jul 2024

Technical Intern

UAE

- Studied VFD operation, including motor speed control, parameter configuration, fault protection, and troubleshooting procedures for industrial automation hardware.
- Practiced PLC ladder logic, digital I/O control, and basic automation sequences while learning how firmware-style control logic interfaces with physical equipment.
- Observed control-system maintenance workflows involving electrical equipment, documentation, component-level troubleshooting, and operational fault analysis.

Crescent Petroleum

Jul 2023 – Aug 2023

IT Intern

UAE

- Collaborated with IT teams on cloud migration and virtual machine tasks, gaining exposure to infrastructure modernization, access control, and system configuration workflows.
- Documented system configurations, access requirements, and support procedures to improve knowledge transfer and troubleshooting workflows.
- Replaced defective components and upgraded office equipment across 10+ workstations, improving reliability and reducing recurring hardware issues.

Crescent Petroleum

Jun 2023 – Jul 2023

Project Management Intern

UAE

- Tracked project milestones, timelines, and deliverables to support planning, execution, monitoring, and short-term risk identification for energy-management related tasks.

Projects

Predictive Temperature-Aware Fan Controller | STM32, Embedded C, PWM, EXTI, UART, LCD, Sensors

2026

- Designed a bare-metal STM32 firmware system that adjusts fan speed using real-time temperature and temperature rate-of-change to improve thermal response and reduce overshoot.
- Implemented PWM motor control, external-interrupt mode switching, UART debugging, LCD status output, buzzer alerts, and modular peripheral drivers for sensor-based control.
- Debugged timing, GPIO, interrupt, and peripheral behavior during hardware bring-up, validating embedded response under changing temperature conditions.

I/O Expander and Encoder Driver Modules | Embedded C, I2C, SPI, GPIO, Rotary Encoders

2026

- Developed embedded driver logic for MCP23008/MCP23S08 I/O expanders to control LEDs, read digital inputs, and validate I2C/SPI bus transactions.
- Built encoder input handling with A/B signal reading, debouncing logic, and scalable GPIO expansion for hardware interface testing.

Custom I2C Master Controller | Verilog, AXI4-Lite, FIFOs, Digital Design

2026

- Implemented an I2C master controller supporting START/STOP conditions, repeated-START, ACK/NACK handling, multi-byte transfers, AXI4-Lite control, and FIFO-based buffering.
- Exposed control/status behavior through memory-mapped registers and debug signals to support transaction validation and hardware/software integration.

IR Communication System | TM4C123GXL, Embedded C, UART, PWM, Interrupts

2025

- Developed a system to transmit, decode, and display NEC protocol commands through a UART terminal using interrupt-driven timing capture and IR waveform generation.

Pipelined RV32I Processor | Verilog, FPGA, Vivado, Computer Architecture

2026

- Designed a 5-stage RV32I-style processor in Verilog with instruction fetch, decode, execute, memory, and writeback stages.
- Implemented register file integration, dual-port memory, hazard detection, forwarding, branch support, load-use stall handling, and assembly-level validation programs.

Embedded Firmware RAG Copilot | Python, FastAPI, Streamlit, Docker, ChromaDB

2026

- Built a local tool that retrieves MCU datasheet context and generates register-level C guidance for peripheral configuration, firmware validation, and embedded debugging workflows.

Technical Skills

Languages: C, C++, Python, Verilog, SystemVerilog, SQL, MATLAB, Java, Bash

Firmware/Embedded: STM32, TM4C123GXL, bare-metal C, GPIO, UART, SPI, I2C, PWM, ADC, timers, interrupts, EXTI, SysTick, sensor interfacing, LCD drivers, embedded debugging, RTOS/HAL concepts

Hardware/FPGA: PCB schematic reading, electronics lab debugging, sensor signal capture, FPGA prototyping, RTL design, AXI4-Lite, FIFOs, Vivado, Vivado Tcl

Tools: Git/GitHub, pytest, STM32CubeProgrammer, Keil uVision, Docker, FastAPI, Streamlit, Jupyter, Pandas, Microsoft Office